

Aditya Kumar

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IT graduate skilled in software development, cybersecurity, and cloud computing - with hands-on experience in full-stack development, ML-based threat detection, and security operations.

EDUCATION

INSTITUTE OF ENGINEERING AND MANAGEMENT

Bachelor of Technology in Information Technology - (CGPA: 9.25)

Trident Public School, Muzaffarpur - Class XII (82%)

Sunshine Prep./High School, Muzaffarpur - Class X (91.4%)

Kolkata

Oct 2022-May 2026

2019-2021

2007-2019

EXPERIENCE

- **NOIDA POWER COMPANY LIMITED (SOC Analyst Intern) (June 2025 - August 2025)** *-Onsite*
 - Used SIEM tools and performed vulnerability assessments for threat detection and incident response.
 - Tested 40+ MDM use cases for policy enforcement, compliance, and remote security management.
- **NIRVEONX (AIML & Cybersecurity Intern) (May 2025 - June 2025)** *-Remote*
 - Led quality-testing initiatives for AI/ML systems, ensuring model accuracy, robustness, and performance.
 - Supported security compliance by identifying control gaps and aligning with ISO 27001 and NIST.

PROJECTS

- **Steganalysis-Based Malware Detection** | Tech Stack: Python, VirusTotal API, Entropy [Live Demo](#) | [GitHub](#)
 - Developed a Python-based tool to **detect hidden malware** in PNG/JPG images by analysing file structure, entropy, and cryptographic hashes without altering the original files.
 - **Automated threat scoring, payload extraction**, and VirusTotal integration - designed for scalable use in SOC environments and future compatibility with **SIEM/EDR systems**.
- **Zero-Knowledge Secure Note Vault** | Tech Stack: Python, FastAPI, Redis, React [Live Demo](#) | [GitHub](#)
 - Built a secure note-sharing web app using client-side AES-256-GCM **encryption**, keeping keys hidden in **URL hash fragments** so the server remains completely blind to the message content.
 - Designed a split-flow API to **stop chat-bot previews** from accidentally burning notes, backed by native Redis TTL self-destruction.
- **ScrapeSentinel-Web Scraping Detection & Mitigation** | Tech Stack: Python, Flask, LightGBM, React
 - Built a **LightGBM - Random Forest ensemble** for **tiered bot detection (T1-T3)** using behavioural and TLS-based features.
 - Developed a **content protection pipeline** with honeytrap tokens, SimHash fingerprinting, and real-time Flask-based attack monitoring.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java.
- **Software Development:** HTML, CSS, JS, Node.js, React, Git, GitHub, MySQL, Spring, Spring Boot.
- **Cloud Security & Services:** AWS (EC2, S3, CloudWatch, VPC, EBS, IAM, Lambda, RDS), Azure, GCP.
- **Penetration Testing:** Nmap, Metasploit, Burp Suite, Privilege Escalation.
- **Machine Learning:** TensorFlow, Scikit-learn.

ACHIEVEMENTS

- Ranked 3rd (2022-2023) and 5th (2024-2025) in IT Department.
- **79th Position in ISCP 2.0 Flipkart CTF challenge.**

CERTIFICATIONS

- **Coursera:** Foundations of Cybersecurity.
- **NPTEL:** Ethical Hacking, The Joy of Computing using Python, Operating Systems Fundamentals.